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Module: 7COM1025-0105-2025 - Programming for Software Engineers

Coursework Report: Fitness and Leisure Centre Booking System

1. System Requirements

This coursework presents the design and implementation of a console-based booking system for a Fitness and Leisure Centre (FLC). The purpose of the system is to manage lesson bookings for members attending various fitness activities such as Yoga, Zumba, Box Fit and Aquacise.

The system allows members to view available lessons, make bookings, change existing bookings and cancel them when necessary. Members can also attend lessons and provide ratings and reviews for the activities they participate in. In addition to these features, the system can generate monthly reports showing lesson attendance and the most popular exercise type based on income generated.

The project was implemented using the Java programming language and follows object-oriented programming principles. Several classes were designed to represent real world entities such as members, lessons and bookings. A UML class diagram was also created to illustrate the structure and relationships of the system.

2. Functional Requirements

The system was designed to support the following core functionalities required by the Fitness and Leisure Centre.

First, users must be able to view the timetable of lessons. This can be done either by selecting a specific day or by searching for lessons by exercise type. This allows members to easily find available classes.

Second, members must be able to book a lesson. Each lesson has a limited capacity, therefore the system ensures that a lesson cannot be booked once it is full. The system also prevents duplicate bookings so that a member cannot book the same lesson more than once.

Third, the system allows members to modify existing bookings. A booking can be changed to a different lesson if the new lesson has available spaces.

Fourth, bookings can also be cancelled. When a booking is cancelled, the available capacity of the lesson is updated so that other members may book it.

Another important feature is lesson attendance. After attending a lesson, members can leave a rating and a short review describing their experience. This information is stored in the system and used when generating reports.

Finally, the system produces two types of reports. The first report shows the attendance and average rating for each lesson during a specific month. The second report identifies the exercise type that generated the highest income during that month.

3. System Design

The system was designed using object-oriented programming concepts such as encapsulation, abstraction and class relationships. Each important concept in the system is represented by a class.

The central class of the system is **FLCSystem**, which manages all the data and operations. This class stores collections of members, lessons and bookings using maps for efficient access.

The **Member** class represents a member of the fitness center. Each member has a unique identifier and a name.

The **Lesson** class represents a fitness lesson. A lesson contains information such as the lesson ID, weekend number, day, time slot, exercise type and price. Each lesson also has a fixed capacity which limits the number of members who can attend.

The **Booking** class represents a reservation made by a member for a particular lesson. It stores the booking ID and maintains references to a Member and a Lesson, along with the booking status.

The **Review** class stores feedback provided by members after attending a lesson. A review includes a rating and a comment.

To simplify the creation of initial data, a helper class called **DataSeeder** was implemented. This class generates sample members and lessons when the system starts.

The system also uses enumeration types for values that have limited options. These include **Day**, **TimeSlot** and **BookingStatus**. Using enumerations improves code clarity and prevents invalid values.

- **3.1 Principles of Design**

  The system is based on important object-oriented design ideas. Encapsulation happens when you keep class attributes private and let people access them only

through public methods. Abstraction is a way to hide implementation details and show only the most important functions.

You can also use separation of concerns by splitting the system into different classes, each of which is in charge of a certain job. For instance, the FLCSystem class takes care of everything, while the Booking and Lesson classes take care of their own logic. This design makes the code easier to read, maintain, and scale.

4. UML Class Diagram

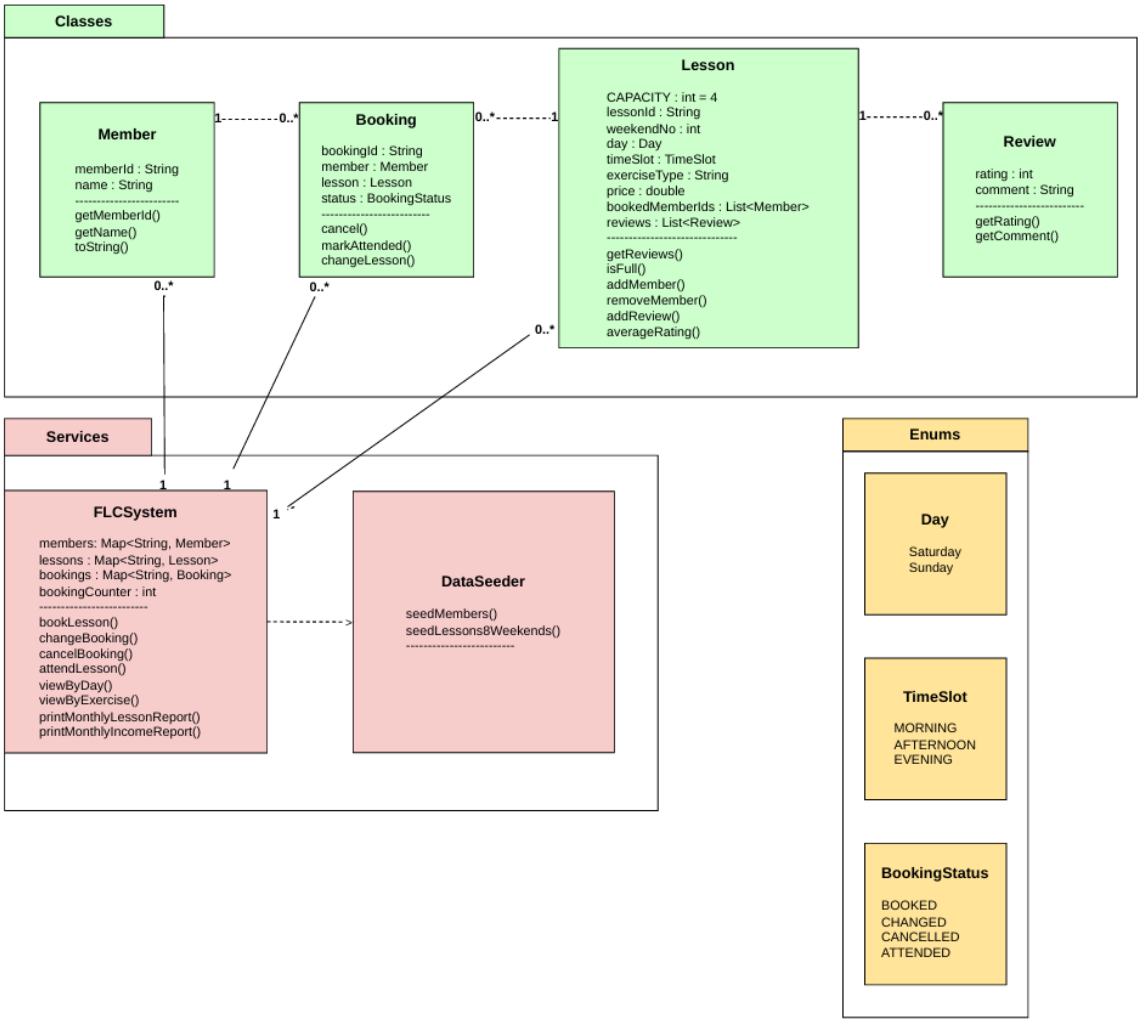
A UML class diagram was created to represent the structure of the system and the relationships between classes.

The diagram shows the main classes including Member, Lesson, Booking, Review and FLCSystem. It also illustrates how these classes interact with each other.

A member can have multiple bookings, while each booking belongs to a single member. Similarly, a lesson can have multiple bookings, but each booking refers to only one lesson. Lessons may also have multiple reviews provided by members after attending.

The FLCSystem class manages collections of members, lessons and bookings, acting as the central controller of the system.

The UML diagram helps visualize the design of the application and ensures that the relationships between classes are clearly defined before implementation.



5. Implementation

The system was implemented using Java and developed using the IntelliJ IDEA development environment. The project was structured as a Maven project, which allows dependencies and build processes to be managed easily.

The implementation follows a modular structure where different responsibilities are assigned to different classes.

The Member class contains simple attributes and methods to access member information. The Lesson class manages lesson details and maintains a list of members who have booked the lesson. It also contains logic to determine whether the lesson is full.

The Booking class represents the connection between a member and a lesson. Methods were implemented to allow bookings to be cancelled, changed or marked as attended.

The Review class stores feedback information provided by members, including a rating and optional comments.

The main system logic is contained in the FLCSystem class. This class provides methods for booking lessons, changing bookings, cancelling bookings, marking attendance and generating reports.

The application is executed through a simple console menu interface which allows the user to choose different operations.

6. Testing

Testing was an important part of the development process to ensure that the system functions correctly.

Unit testing was implemented using **JUnit 5**. Several test cases were written to verify the behaviour of the booking system.

For example, one test checks that a lesson can be successfully booked when space is available. Another test verifies that duplicate bookings are not allowed for the same member and lesson.

Additional tests ensure that the lesson capacity limit is enforced and that a fifth booking cannot be made when the lesson capacity is four. Tests were also created to verify booking cancellation and lesson attendance functionality.

These tests help confirm that the system behaves as expected and that important constraints such as capacity limits and duplicate bookings are properly handled.

7. Refactoring

The code was refactored during development to make it easier to read and understand. Some methods were made easier to understand, and the work was spread out more evenly among classes. This helped cut down on code duplication and made the system easier to keep up with.

Refactoring also made sure that the design still followed object-oriented principles.

8. Version Control

Git was utilized for version control during development. Commits were made on a regular basis to monitor system modifications and development. This made it possible to handle code more effectively and guaranteed that earlier versions could be restored if necessary.

Commit history screenshots are provided to show constant development.

Commits on Mar 6, 2026

README file update according to work SZAEEMHASSAN committed on Mar 6	ac40ed1

Commits on Mar 5, 2026

lesson class and review class are created SZAEEMHASSAN committed on Mar 5	3b943ea

Commits on Mar 3, 2026

Readme file is updated according to work SZAEEMHASSAN committed on Mar 3	d1b9949
Day, BookingStatus, TimeSlot enums and Member class are created SZAEEMHASSAN committed on Mar 3	099e0c3

Commits on Mar 2, 2026

Resolve README merge conflict SZAEEMHASSAN committed on Mar 2	b1b042e
Maven project structure created and base packages added SZAEEMHASSAN committed on Mar 2	f05cbe7
Initial Project Setup SZAEEMHASSAN committed on Mar 2	bad6cb9
Initial commit SZAEEMHASSAN authored on Mar 2	Verified 7421ad0

The commit history shows the gradual implementation of system components, including class creation and project structure setup.

9. Conclusion

In conclusion, all necessary features are successfully met by the Fitness and Leisure Center booking system. Members may effectively manage reservations thanks to the system, which also offers helpful reports for analysis.

The system's organization and maintainability are guaranteed by the application of object-oriented design concepts. A graphical user interface and database integration for long-term storage are two possible future enhancements.